Learning Healthy Anatomy to Find Tumors: Comparing VAEs and GANs for Unsupervised Anomaly Detection in Brain MRI (Midway Report)

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Abstract

Supervised tumor segmentation needs thousands of expert-drawn outlines and only recognizes the diseases it was trained on. We instead train generative models only on *healthy* brain MRI and flag the regions a model cannot reconstruct. We compare a variational autoencoder (VAE), f-AnoGAN (a Wasserstein GAN with an encoder) and a plain autoencoder (AE) under one shared pipeline. Models are trained on 462 healthy IXI subjects (T2) and tested on 436 glioma patients from the Medical Segmentation Decathlon. So far we have built the full pipeline, including our own PyTorch affine registration because the standard tools do not run on our hardware, and trained and evaluated all three models at 64×64. The VAE is best (subject Dice 0.51, pixel AUROC 0.89), f-AnoGAN is close (Dice 0.47) with the fewest complete misses, and the plain AE does not beat a simple "bright on T2" intensity baseline on Dice (0.38 vs. 0.39). Our main challenges were registration failures on about 30% of scans and slow CPU training. Next we will move to the proposal's 128×128 resolution, add seeds and confidence intervals, and sweep the VAE's KL weight and latent size.

1. Introduction

Most deep learning tools for medical imaging are supervised: they learn from scans in which a radiologist has outlined every lesion. Such labels are slow and expensive to produce, and a supervised model only recognizes the abnormalities it was shown. Healthy scans, by contrast, are plentiful.

We ask whether a generative model that has only ever seen healthy brains can detect and localize brain tumors, and whether a VAE or a GAN does this better. The idea is to learn the distribution of healthy anatomy, $p(\mathbf{x})$. When the model is shown a scan with a tumor, it "repairs" the image toward the nearest healthy-looking brain $\hat{\mathbf{x}}$, and the residual $\mathbf{x} - \hat{\mathbf{x}}$ highlights what does not belong. Because no tumor labels are used for training, this *unsupervised anomaly detection* could in principle flag abnormalities nobody trained it to find. This makes it attractive as a "second reader" for radiologists.

Generative models are central to this approach. A plain autoencoder only learns to copy its input, and with enough capacity it copies tumors too. A VAE (latent prior) and a GAN (realism critic) both restrict reconstructions to images that look like the training distribution. VAEs are stable but blurry, so fine edges cause false alarms. GANs are sharp but harder to train and may not cover all of healthy anatomy. Published comparisons disagree, partly because they use different private data and preprocessing (Baur et al., 2021). Our contribution is a controlled comparison on public data, with identical preprocessing and post-processing for every model.

2. Related Work

Generative models. VAEs (Kingma & Welling, 2014) maximize a lower bound on the data likelihood with an encoder–decoder pair and a Gaussian latent prior. GANs (Goodfellow et al., 2014) train a generator against a discriminator. DCGAN (Radford et al., 2016) made GANs workable on images with strided convolutions and batch normalization, and the Wasserstein GAN with gradient penalty (WGAN-GP) (Gulrajani et al., 2017) stabilized training further. We use DCGAN-style layers for all models so that differences come from the training objective, not the architecture.

GANs for anomaly detection. AnoGAN (Schlegl et al., 2017) was the first to use a GAN trained on healthy images to localize anomalies in medical images. It finds the closest healthy image by optimizing the latent code separately for every test image, which is slow. f-AnoGAN (Schlegl et al., 2019) replaces this search with an encoder trained after the WGAN-GP, giving a single forward pass. We adopt f-AnoGAN as our GAN representative.

Autoencoders on brain MRI. Baur et al. (2018) combined a VAE with an adversarial loss to segment multiple-sclerosis lesions without labels. Chen & Konukoglu (2018) constrained the latent space of an adversarial autoencoder for brain lesions, and Zimmerer et al. (2019) showed that VAE gradients and the KL term also carry localization informa-

tion. The comparative study of Baur et al. (2021) benchmarked many AE and GAN variants on shared data. It found that results depend heavily on preprocessing and post-processing, and that no model family clearly wins. Meissen et al. (2021) went further: on FLAIR scans a simple intensity threshold matched or beat these deep models, because tumors are simply hyperintense. More recent work uses diffusion models (Pinaya et al., 2022), but VAEs and GANs remain the standard reference points.

Our position. Following these findings, we (i) use only public data, IXI (Imperial College London) and MSD Task01 (Antonelli et al., 2022; Menze et al., 2015); (ii) share every preprocessing and scoring step between models; (iii) include a plain AE to test whether *generative* modelling helps; and (iv) include the intensity baseline that Meissen et al. (2021) found to be competitive.

3. Methods (Implemented)

All code is implemented in PyTorch, version-controlled in git and covered by 19 unit and end-to-end tests. Every model and baseline below has been trained and evaluated.

3.1. Preprocessing pipeline

IXI and MSD come from different scanners and were processed differently, so a model could flag "different hospital" instead of "tumor". We therefore apply one pipeline to every scan: (1) reorient to RAS and roughly rescale intensities; (2) affinely register (12 parameters) to the MNI152 2009a template (Fonov et al., 2011); (3) skull-strip with the template's brain mask (MSD scans are already skull-stripped, so we also intersect with their own brain outline); (4) divide by the 99th percentile of brain intensities and clip to $[0, 1]$; (5) keep axial slices whose brain area is at least 30% of the maximum, taking every second level, crop around the brain and resize to 128×128 by area averaging. The 64×64 inputs are pooled from these slices.

Challenge 1: no registration library. Our team develops on Windows laptops with ARM processors (Snapdragon X Elite). SimpleITK, ANTs and dipy have no builds for this platform, and neither does any dedicated skull-stripping tool. We therefore wrote affine registration ourselves in PyTorch. It resamples with `grid_sample`, maximizes normalized cross-correlation (NCC) with Adam, works coarse to fine (4 mm, then 2 mm), and uses a random subset of 40k voxels per step for speed. A unit test confirms it recovers a known affine transform to under 1 mm. Nilearn ships only a T1 template, so we built a T2-like template from its gray- and white-matter probability maps (CSF $= 1.0$, GM $= 0.55$, WM $= 0.35$, matching T2 contrast).

Challenge 2: registration failures. In our first full run, about 30% of IXI scans registered badly (NCC ≈ 0, with the eyes inside the brain mask). Head tilt and a poor centre-of-mass start trapped the optimizer in local minima, and trying several starting shifts did not help. What worked was a coarse grid search, at 6 mm, over pitch ($\pm 25°$), anterior and vertical shifts (± 20 and ± 40 mm) and scale (0.8–1.0). The three best starting points are each refined, and the best of them initializes the main optimization. After this change all 578 IXI and 484 MSD scans pass a visual check. Median NCC is 0.47 for IXI and 0.25 for MSD. We confirmed that the lowest MSD scores belong to correctly aligned scans with very large tumors, not to failed registrations. Preprocessing takes about 25 s per scan, or 40–45 minutes per dataset with 6 workers.

3.2. Models

All models share DCGAN-style networks. The encoder applies four stride-2 4×4 convolutions ($1 \to 32 \to 64 \to 128 \to 256$ channels, LeakyReLU 0.2, batch norm), turning a 64×64 image into a $256 \times 4 \times 4$ feature map, followed by a linear layer. The decoder mirrors this with transposed convolutions and a sigmoid output. The latent size is $d = 128$, and weights follow the DCGAN initialization.

AE (2.44M parameters) minimizes $\|\mathbf{x} - D(E(\mathbf{x}))\|_2^2$.

VAE (2.96M) minimizes the negative ELBO with a Gaussian decoder:

$$\mathcal{L}_{\text{VAE}} = \|\mathbf{x} - D(\mathbf{z})\|_2^2 + \beta \,\mathrm{KL}\big(q(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, I)\big), \quad (1)$$

with $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$ (the reparameterization trick) and $\beta = 1$. At test time we decode the posterior mean.

f-AnoGAN (3.13M) is trained in two stages. Stage 1 is a WGAN-GP: the generator G is the decoder architecture and the critic C is the encoder architecture with layer normalization instead of batch normalization, because the gradient penalty is computed per sample. The critic minimizes $\mathbb{E}[C(G(\mathbf{z}))] - \mathbb{E}[C(\mathbf{x})] + \lambda\, \mathbb{E}[(\|\nabla_{\tilde{\mathbf{x}}} C(\tilde{\mathbf{x}})\|_2 - 1)^2]$ with $\lambda = 10$ and $n_{\text{critic}} = 5$. Stage 2 freezes G and C and trains an encoder E with the izi$_f$ loss $\|\mathbf{x} - G(E(\mathbf{x}))\|^2 + \kappa \|f(\mathbf{x}) - f(G(E(\mathbf{x})))\|^2$, where f denotes the critic's last feature map and $\kappa = 1$.

3.3. Anomaly scoring

Gliomas and the swelling around them are bright on T2, so our main anomaly map keeps only the *positive* residual:

$$A = \big(g_\sigma * \max(\mathbf{x} - \hat{\mathbf{x}},\, 0)\big) \odot M_\ominus, \quad (2)$$

where g_σ is a Gaussian ($\sigma = 1$ pixel) and $M_\ominus$ is the brain mask eroded by 2 pixels, because reconstructions are least reliable at the brain border. A pixel is labelled tumor if $A > \tau$. The threshold τ maximizes Dice on a held-out

MSD validation set. A slice's score is the mean of its top 1% of anomaly values. We also report the plain absolute residual $|\mathbf{x} - \hat{\mathbf{x}}|$ as an ablation. The *intensity baseline* uses $\hat{\mathbf{x}} = \mathbf{0}$, so it detects "bright = tumor".

Challenge 3: compute. All results so far were run without a GPU, on a Windows laptop with an ARM processor. On the native ARM64 PyTorch build the AE trained at only 257 images/s. The x64 build, running under Windows emulation, uses optimized oneDNN kernels and reached 670 images/s. We therefore train with the x64 build, running all three models in parallel with 4 threads each. This compute limit is why our first results use the proposal's 64×64 "fast experiment" setting.

4. Experiments

4.1. Data and setup

Training data: IXI (Imperial College London), 578 healthy adults, T2-weighted, split 80/10/10 by subject into 462 training, 58 validation and 58 test subjects (26,796 training slices). **Test data:** MSD Task01 BrainTumour (Antonelli et al., 2022), 484 glioma patients from BraTS (Menze et al., 2015). We use the T2 channel and merge all tumor labels into one "whole tumor" mask. 48 subjects (10%) are used only to choose the threshold τ, and 436 subjects (25,288 slices) are the test set. Tumor pixels make up 9.2% of MSD test brain pixels. **Training:** Adam with learning rate $2{\cdot}10^{-4}$ and batch size 64, random left–right flips, 40 epochs for AE and VAE. f-AnoGAN uses 60 WGAN-GP epochs (learning rate 10^{-4}, $\beta = (0, 0.9)$) followed by 30 encoder epochs. For each model we keep the checkpoint with the lowest healthy validation MSE. Wall-clock time on CPU was 2.9 h each for AE and VAE and 6.3 h for f-AnoGAN (5.4 h of which was the GAN stage). **Metrics** follow the proposal: Dice, pixel AUROC and AUPRC, two slice AUROCs, and the false-positive rate on healthy IXI test pixels.

4.2. Preliminary results

Training behaved as expected (Figure 2). The AE reaches by far the lowest healthy reconstruction error ($2.1 \cdot 10^{-3}$, vs. $4.7 \cdot 10^{-3}$ for the VAE and $5.6 \cdot 10^{-3}$ for f-AnoGAN). The Wasserstein estimate falls steadily from 35.5 to 1.5 without collapse, and GAN samples show realistic anatomy.

Generative models win, and the AE is the cautionary tale (Table 1). The VAE is best on every pixel-level metric. It improves subject Dice by $+0.13$ over both the AE and the intensity baseline and flags $3.7\times$ fewer healthy pixels. f-AnoGAN comes second on pixel metrics and ties the VAE at slice level (AUROC 0.93 vs. healthy IXI). The plain AE, despite reconstructing healthy brains best, has the *lowest* pixel AUROC (0.80). Figure 1 shows why: it reconstructs part of the tumor too, so the residual misses it. Good reconstruction is not the goal; reconstructing *only* healthy anatomy is. This confirms the motivation for constraining the latent space. The intensity baseline is strong at pixel level (AUROC 0.85), in line with Meissen et al. (2021), but poor at slice level (0.77), since many bright healthy structures such as ventricles and CSF also trigger it.

Post-processing matters. With the absolute residual instead of the positive one, every model loses Dice (VAE $0.514 \rightarrow 0.484$, f-AnoGAN $0.474 \rightarrow 0.435$, AE $0.380 \rightarrow 0.334$). The healthy FPR of f-AnoGAN doubles ($1.3\% \rightarrow 2.6\%$), because dark mismatches at fine folds are now counted.

Small tumors are the main failure mode (Figure 3). Splitting test subjects into thirds by tumor size, VAE Dice rises from 0.34 (small) to 0.65 (large), and every model shows this pattern. The groups also separate the two generative models. The VAE leads on large tumors (0.65 vs. 0.57), but f-AnoGAN has the fewest near-complete misses: 8.7% of subjects have Dice < 0.2, compared with 11.7% for the VAE, 14.7% for the AE and 28.2% for the baseline. This suggests that the VAE's blur hides small lesions less than we expected, while f-AnoGAN's scattered false positives mostly hurt on large tumors, where Dice is dominated by boundary quality.

Caveats. Each number comes from a single run (seed 42), so differences of a few Dice points are not yet established. Our template-mask skull stripping leaves a thin rim of skull on some IXI scans, and MSD scans were skull-stripped and resampled differently, so part of every residual reflects domain shift.

5. Conclusion and Updated Plan

Progress. The full pipeline is working end to end: data download, custom registration and preprocessing, three models, a baseline, evaluation and figures. Our first results already answer the proposal's question at 64×64. Both generative models clearly beat a plain AE, the VAE is best on average and f-AnoGAN is the most consistent. **Changes from the proposal:** we had to write our own registration and T2 template because no standard tool runs on our hardware, and we added the intensity baseline and the positive-residual scoring after reviewing Meissen et al. (2021).

Next steps (by priority). (1) *Robustness:* retrain each model with 3 seeds, report bootstrap confidence intervals over subjects, and run a paired Wilcoxon test of VAE vs. f-AnoGAN Dice. (2) *Resolution:* train at 128×128, the proposal's main setting. This costs about $4\times$ more compute (an estimated 12 h for AE and VAE and over 24 h for f-AnoGAN on the laptop CPU), so we plan to migrate training to a Google Colab GPU runtime for much faster runs. (3) *VAE trade-off:* sweep $\beta \in \{0.5, 1, 2\}$ and latent

Table 1. Test results at 64×64 (positive residual). Dice is computed per subject (mean $\pm$ sd over 436 MSD test subjects) and over all test pixels pooled (global). Slice AUROC (MSD) separates tumor slices from tumor-free slices of MSD patients. Slice AUROC (vs. IXI) separates MSD tumor slices from healthy IXI test slices. FPR is the fraction of healthy IXI test brain pixels flagged at the chosen threshold. Best value per column in bold.

Model	Dice (subject)	Dice (global)	Pixel AUROC	Pixel AUPRC	Slice AUROC (MSD)	Slice AUROC (vs. IXI)	Healthy FPR
Intensity baseline	0.394 ± 0.243	0.431	0.854	0.351	0.779	0.772	4.52%
AE	0.380 ± 0.161	0.420	0.796	0.402	**0.869**	0.865	4.01%
VAE	$\mathbf{0.514 \pm 0.227}$	**0.566**	**0.894**	**0.575**	0.867	0.929	**1.21%**
f-AnoGAN	0.474 ± 0.183	0.517	0.855	0.514	0.867	**0.931**	1.31%

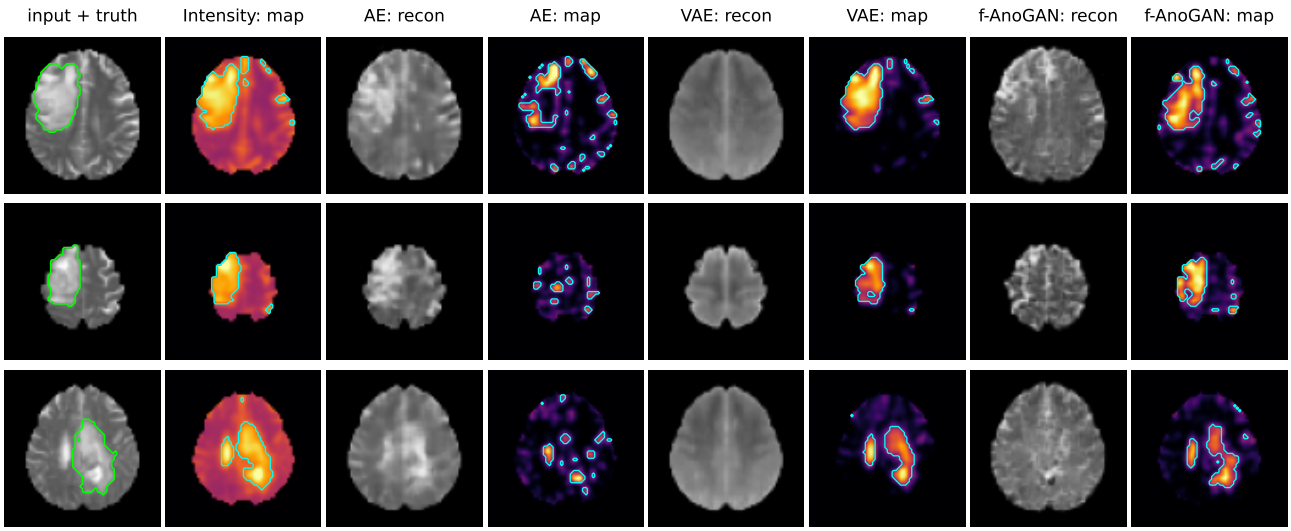


Figure 1. MSD test slices (rows) with each model's healthy reconstruction and anomaly map. Green: expert whole-tumor outline. Cyan: predicted tumor at the threshold chosen on validation. The VAE's blurry reconstruction leaves one compact anomaly. f-AnoGAN is sharper but adds scattered false positives where folds are misplaced. The AE partly reconstructs the tumor itself, which weakens its map.

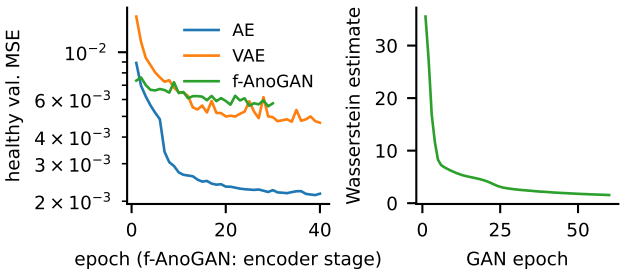


Figure 2. Left: reconstruction error on healthy IXI validation slices. Right: the f-AnoGAN critic's Wasserstein estimate during GAN training.

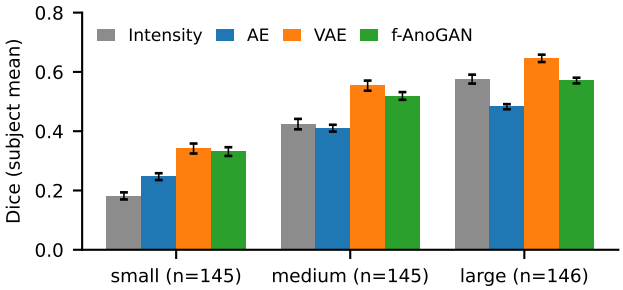


Figure 3. Subject Dice by tumor-size tertile (whole-tumor pixels summed over a subject's 64×64 slices). Error bars show the standard error.

size $d \in \{128, 256\}$ to measure how blur trades sharpness against false alarms. (4) *Error analysis:* break down results by MSD tumor sub-region, and test whether running HD-BET (Isensee et al., 2019) skull stripping on Colab reduces border false positives.

Timeline. *Weeks of Nov 9 and Nov 16:* seed repeats and confidence intervals at 64×64; migrate to a Colab GPU runtime; start the 128×128 runs. *Week of Nov 23:* β and latent-size sweep; sub-region and tumor-size analysis. *Week of Nov 30:* HD-BET ablation; freeze experiments. *Week of Dec 7:* final report.

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